Enhancing Business Intelligence for Contract Management: Semantic-Aware LLM-Based Analytics for Decision Support

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Abstract: Business Intelligence (BI) for Contract Management requires integrating and interpreting heterogeneous information from structured systems and unstructured contractual documents. Recent advances in Large Language Models (LLMs) have enabled the development of BI systems that combine techniques such as Retrieval-Augmented Generation (RAG), Text-to-SQL, and agent-based orchestration to support complex queries over multiple data sources. However, these approaches remain fundamentally data-driven, relying on retrieval and probabilistic generation without explicit semantic grounding. As a result, they face limitations in handling semantically complex queries involving obligations, rights, and contractual responsibilities, where meaning is distributed and implicitly defined. In this paper, we propose a semantic-aware architecture for LLM-based BI in contract management that introduces a formal Semantic Layer grounded on ontology-enriched knowledge models. We present a Semantic Onboarding methodology that transforms heterogeneous contractual artifacts into structured, executable representations by combining LLM-based extraction, deterministic schema construction, and ontology alignment. These models enable direct querying of contractual semantics through logic-based reasoning, supporting more consistent, complete, and explainable decision-support capabilities. We evaluate the proposed approach against a retrieval-based baseline and show that semantic grounding significantly improves the system's ability to answer semantically grounded questions. In particular, the results demonstrate gains in semantic completeness, interpretability, and consistency when analyzing contractual obligations and enforcement relations. The proposed approach represents a shift from data integration toward knowledge construction, providing a foundation for semantically grounded decision-support systems in contract management and related domains.

1 Introduction

Effective decision-making in contract management requires the integration and interpretation of heterogeneous information sources, including structured data from contract management systems and unstructured data from contractual documents. In domains such as IT service contracts, relevant information is often distributed across clauses, annexes, and operational records, making it difficult to obtain consistent and actionable insights. Business Intelligence (BI) systems have traditionally addressed this challenge by enabling aggregation, visualization, and analysis of structured data. However, their ability to incorporate unstructured contractual knowledge remains limited.

Recent advances in Large Language Models (LLMs) have significantly expanded the capabilities of BI systems by enabling the integration of heterogeneous data sources through natural language interfaces. Techniques such as Retrieval-Augmented Generation (RAG), Text-to-SQL, and agent-based orchestration allow LLM-based systems to retrieve and synthesize information from both documents and databases. In a previous work (Seabra et al., 2025b), we presented an LLM-based BI system for contract management that combines these techniques to provide actionable insights over contractual data. The system demonstrated strong performance in answering complex queries and supporting decision-making tasks by dynamically integrating multiple data sources.

Despite these advances, current LLM-based BI systems remain fundamentally data-driven. Their op-

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eration is largely based on retrieving relevant fragments of information and synthesizing responses through probabilistic generation. As a result, they lack explicit semantic grounding, making it difficult to ensure consistency, completeness, and interpretability in scenarios that require structured reasoning. Consider, for example, the question *What is the cybersecurity criticality level of the contract?*. Answering this question requires combining contractual obligations, access privileges, data sensitivity, operational responsibilities, and organizational criticality criteria distributed across multiple contractual and governance artifacts. This example highlights a fundamental gap between data integration and knowledge construction in LLM-based systems. Although existing approaches focus on accessing and combining information, they do not explicitly construct the semantic structures necessary for reasoning over such contractual knowledge. In contract management, where meaning is often distributed across multiple clauses and expressed through normative relations, this limitation becomes particularly critical.

To address this challenge, this paper proposes a semantic-aware architecture for contract management that introduces an explicit *Semantic Layer* grounded on ontology-enriched formal knowledge models. Instead of relying solely on retrieval and generation, the proposed approach constructs formal representations of contractual knowledge through a Semantic Onboarding methodology. This methodology transforms heterogeneous contractual artifacts into structured, executable, and ontology-aligned models that can serve as a semantic substrate for decision support.

The main contributions of this paper are (i) a Semantic-Aware architecture for LLM-based Business Intelligence in contract management, centered on a formal Semantic Layer that supports semantically grounded reasoning, (ii) a Semantic Onboarding methodology that transforms heterogeneous contract artifacts into ontology-enriched, executable knowledge models, combining LLM-based extraction, deterministic schema construction, and ontology alignment, and (iii) an evaluation based on a real use case of how ontology-aligned formal representations enable direct querying of contractual semantics, including obligations, rights, and enforcement relations, through logic-based reasoning.

The remainder of this paper is organized as follows. Section 2 presents the conceptual foundations of formal knowledge representation, ontologies, and LLM-based knowledge extraction. Section 3 discusses related work in knowledge graph construction, ontology-based modeling, and logic-based formalization. Section 4 introduces the Semantic Onboarding methodology. Section 5 describes the proposed system architecture. Section 6 presents the evaluation and comparative analysis. Finally, Section 7 concludes the paper and outlines directions for future work.

2 Background

This section introduces the main concepts that support the proposed approach. We first discuss formal knowledge representation as a foundation for machine-interpretable reasoning. We then present the role of ontologies in conceptual modeling, followed by the distinction between TBox and ABox in knowledge bases. Finally, we review recent advances in the use of Large Language Models (LLMs) for knowledge extraction and structuring.

2.1 Formal Knowledge Representation

Formal knowledge representation aims to encode domain information in a form that can be interpreted and processed by computational systems. It enables the explicit modeling of entities, relationships, and constraints, allowing automated reasoning and semantic querying over structured knowledge bases.

Logic-based representations have long been used to model domain knowledge in machine-interpretable form. In such approaches, knowledge is expressed as sets of logical statements that can be evaluated through reasoning mechanisms. Description logics and rule-based systems provide formal semantics to represent domain concepts and relationships while supporting inference over implicit knowledge (Baader et al., 2017; Li et al., 2022). These representations form the basis of many current semantic systems, including knowledge graphs and ontology-driven information systems.

By structuring domain knowledge as formal models, such systems enable reasoning tasks such as classification, constraint checking, and semantic querying. This capability is particularly relevant for decision-support systems, where deriving consistent and interpretable conclusions from heterogeneous information sources is essential. As a result, formal knowledge representations play a central role in data integration platforms, knowledge-driven applications, and intelligent information systems.

2.2 Ontologies and Conceptual Modeling

Ontologies provide a formal and explicit specification of the conceptual structure of a domain. They define the vocabulary used to describe domain entities, as well as the relationships and constraints that connect them. In conceptual modeling and knowledge engineering, ontologies serve as semantic frameworks that support knowledge sharing, interoperability, and reasoning (Li et al., 2022).

In ontology-based systems, concepts are typically organized in hierarchical structures and connected through semantic relations. These structures enable the representation of domain knowledge in a way that is both human-understandable and machine-interpretable. Ontologies are therefore widely used in the construction of knowledge graphs, semantic information systems, and intelligent decision-support platforms. Recent work has emphasized the role of ontologies in integrating heterogeneous data sources and guiding knowledge construction pipelines. In knowledge graph construction, for example, ontologies often provide the conceptual schema that governs the extraction and integration of entities and relations from raw data (Huang et al., 2024). By grounding extracted information in a shared conceptual model, ontology-guided systems can ensure semantic consistency across heterogeneous sources.

Beyond integration, ontologies also play a critical role in enabling semantic grounding. By providing a stable conceptual reference, they allow domain-specific data to be interpreted within a broader semantic framework, supporting more consistent reasoning and decision-making processes.

2.3 TBox and ABox in Knowledge Bases

Knowledge bases in ontology-driven systems are typically divided into two components: the terminological component (TBox) and the assertional component (ABox) (Baader, 2003). The TBox defines the conceptual schema of the domain, including classes, relationships, and logical constraints. In contrast, the ABox contains instance-level assertions describing specific entities and their relationships.

The TBox captures the structural knowledge of a domain, while the ABox represents concrete facts about individual instances (Bienvenu, 2022). Reasoning systems can use the TBox to infer additional relationships within the ABox, enabling tasks such as instance classification, consistency checking, and query answering. This separation between schema-level and instance-level knowledge is central to ontology-based knowledge representation. It allows knowledge bases to maintain a stable conceptual structure while continuously incorporating new instance-level information. In many knowledge construction pipelines, the TBox serves as a schema that guides the generation or extraction of ABox facts from heterogeneous data sources.

This distinction is particularly relevant in systems that rely on automated knowledge construction, as it enables a clear separation between conceptual modeling and data instantiation, supporting both consistency and scalability.

2.4 Large Language Models for Knowledge Extraction

Large Language Models (LLMs) have recently emerged as powerful tools for extracting and structuring knowledge from unstructured and semi-structured data. By leveraging their contextual understanding capabilities, LLMs can identify entities, relations, and events in heterogeneous sources, facilitating the construction of structured knowledge representations (Zhu et al., 2024).

Recent research has explored the use of LLMs for knowledge graph construction and semantic information extraction. These approaches employ LLMs to transform natural language artifacts into structured representations, such as triples or logical facts, that can be incorporated into knowledge bases (Liang et al., 2024). This paradigm represents a shift from traditional rule-based extraction pipelines toward generative approaches capable of integrating heterogeneous information sources. Furthermore, ontology-guided LLM pipelines have been proposed to improve the semantic consistency of extracted knowledge. By grounding generation processes in predefined ontological structures, these approaches constrain model outputs and reduce inconsistencies during knowledge construction (Feng et al., 2024).

These developments highlight the growing role of LLMs as assistants for knowledge engineering tasks, including ontology construction, schema induction, and instance population. However, the representations produced by LLMs are typically derived from contextual interpretation rather than grounded in an explicit semantic framework. As a result, although they capture relevant information, they often lack well-defined conceptual structures and formal semantics that support consistent interpretation and reasoning. This limitation becomes particularly evident in domains where meaning depends on implicit relationships, constraints, and normative structures, such as contract management.

3 Related Work

Research on transforming heterogeneous information artifacts into structured and executable knowledge representations spans several areas, including knowledge graph construction, ontology-based knowledge engineering, and logic-based knowledge formalization. In recent years, LLMs have significantly influenced these directions by enabling scalable extraction and structuring of information from natural language sources.

3.1 LLM-Based Knowledge Extraction and Knowledge Graph Construction

Several approaches leverage LLMs to construct knowledge graphs directly from textual artifacts, identifying entities and relationships that can be organized into graph structures (Zhu et al., 2024). These methods have demonstrated promising results in automatically generating triples and relational structures from large document collections. More recent work (Pan et al., 2023) explores LLM-assisted pipelines that jointly address entity recognition, relation extraction, and schema induction. These approaches often combine neural extraction methods with symbolic representations to improve interpretability and reasoning capabilities.

In parallel, Retrieval-Augmented Generation (RAG) architectures have been widely adopted to improve the relevance of LLM-generated responses (Lewis et al., 2020). These approaches retrieve text segments based on similarity and use them as context for generation. However, similarity-based retrieval does not necessarily imply relevance to the user's intent, particularly in complex analytical scenarios (Seabra et al., 2025a). Extensions such as GraphRAG incorporate structured representations such as knowledge graphs to enhance multi-hop reasoning (Knollmeyer et al., 2025). While these methods improve contextual grounding, they remain fundamentally based on similarity-driven retrieval and stochastic generation.

3.2 Ontology-Guided Knowledge Construction

Another emerging line of research explores integrating LLMs with ontology-driven pipelines for knowledge graph population. In these systems, ontological schemas are used to constrain the extraction process, improving semantic consistency and reducing structural inconsistencies in automatically generated graphs (Zhao et al., 2025a).

Ontology-based knowledge engineering has long been used to structure domain knowledge and enable semantic interoperability across heterogeneous information sources. Ontologies provide explicit conceptual models that define entities, relations, and constraints within a domain (Guarino et al., 2009). In knowledge graph construction pipelines, they often serve as schemas that guide the extraction and integration of heterogeneous data sources (Keet, 2023).

In regulatory and legal domains, ontology engineering methodologies have been applied to support formal reasoning and compliance verification. For example, privacy and regulatory ontologies (Palmirani et al., 2018) have been developed to represent legal concepts and support automated reasoning. These approaches demonstrate the potential of ontological models to structure complex normative knowledge. However, most ontology-based approaches assume that instance-level knowledge is already available or manually curated. In contrast, our work focuses on the earlier stage of knowledge construction, where instance-level representations are automatically generated from heterogeneous artifacts and subsequently aligned with ontological models.

3.3 Logic-Based Formalization and Executable Knowledge

Another important line of research concerns the transformation of natural-language artifacts into executable logical representations. Logic programming languages, such as Prolog, have been widely used to encode formal knowledge models that support automated reasoning. In the legal domain, several systems have explored logic-based representations to formalize statutory rules and regulatory frameworks. For example, the PROLEG system provides a Prolog-based framework for representing legal rules, exceptions, and reasoning structures in an executable form (Satoh, 2023).

Recent work has also investigated integrating LLMs with logic-based knowledge representation frameworks. In particular, human-in-the-loop workflows have been proposed to transform legal provisions into executable rules by combining LLM-based extraction with expert validation (Zin et al., 2026). Additionally, hybrid architectures incorporating structured reasoning into retrieval-based systems have been explored (Zhao et al., 2025b). Despite these advances, most approaches either focus on improving retrieval-based reasoning or generating structured representations that are not directly executable. Fully deterministic, logic-based knowledge models derived automatically from heterogeneous ar-

tifacts remain underexplored.

3.4 Positioning of This Work

While existing approaches have made significant progress in knowledge extraction, retrieval augmentation, ontology-guided modeling, and logic-based formalization, they typically address these aspects in isolation. Retrieval-based systems improve access to information but lack explicit semantic structure. Ontology-based approaches provide conceptual rigor, but often depend on manually curated data. Logic-based systems enable reasoning, but generally assume that formal representations are already available. This work bridges these gaps by proposing a unified approach that combines:

- LLM-based extraction from heterogeneous artifacts;
- deterministic schema construction (TBox induction);
- automated generation of instance-level representations (ABox);
- alignment with ontological structures;
- and executable logic-based knowledge models.

In contrast to prior work, the proposed approach produces semantically grounded, executable knowledge representations that can serve as a foundation for decision-support systems. By integrating these components into a single pipeline, this work advances the state of the art from retrieval-based and graph-based representations toward fully semantic and logic-driven knowledge construction.

4 Methodology

The Semantic Onboarding methodology is responsible for transforming heterogeneous contract-related artifacts into ontology-enriched formal knowledge models that can serve as semantic grounding for decision-support applications. Its purpose is not merely to extract information from documents, but to construct formal and reusable semantic representations capable of supporting consistent interpretation, executable reasoning, and contract-oriented analytics. As illustrated in Figure 1, the methodology is organized into a sequence of interconnected phases that combine large language models, deterministic structuring procedures, ontology alignment, and human validation.

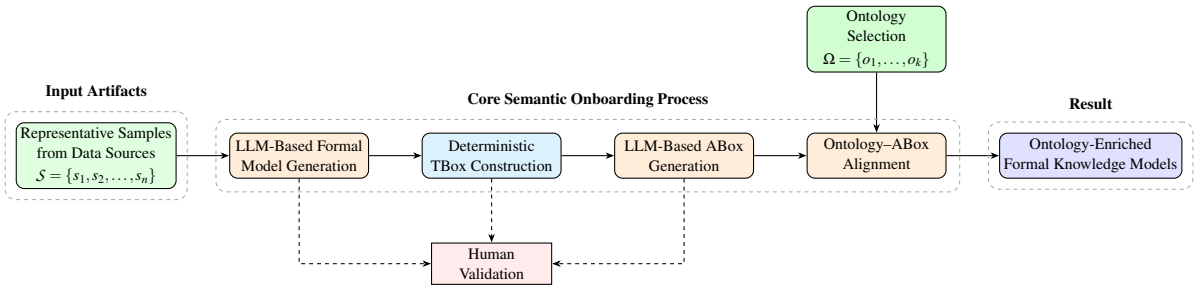
The process begins with the selection of *representative samples from data sources*. Rather than directly formalizing all available contractual artifacts, the methodology starts from a curated set of samples considered representative of the domain structures to be modeled. These samples may include contract documents, annexes, terms of reference, and other related artifacts that express relevant contractual semantics. The purpose of this phase is to provide a sufficiently rich and representative basis for identifying recurring contractual entities, relations, and structures that will later support the construction of formal models. In methodological terms, this initial sampling stage establishes the semantic scope of the onboarding process.

The second phase is *LLM-based formal model generation*. In this phase, a large language model is used to transform representative contractual samples into an initial formal representation. The role of the LLM here is constructive rather than merely extractive: it is expected to identify relevant contractual elements and organize them into a formal structure that captures the semantics of the source artifacts. This phase serves as the first bridge between natural-language contractual material and machine-processable representations. Because contractual documents are often heterogeneous in style, structure, and granularity, the use of an LLM provides flexibility in interpreting and restructuring the content into a more regular formal model.

Once an initial formal representation is obtained, the methodology proceeds to *deterministic TBox construction*. This phase is responsible for consolidating the conceptual schema induced from the formal models generated in the previous step. Although the LLM provides an initial formalization, this phase systematizes the conceptual backbone of the resulting knowledge structure by identifying stable categories, relations, and schema-level regularities. In other words, the methodology distinguishes between schema-level knowledge and instance-level knowledge, and this step explicitly builds the terminological structure that will constrain and guide the next stages. The deterministic nature of this phase is important because it reduces variability and provides a stable conceptual basis for the subsequent generation of formal instances.

The fourth phase is *LLM-based ABox generation*. Once the TBox has been established, the methodology returns to the contractual artifacts and generates instance-level representations according to the schema previously defined. This step ensures that the resulting formal models are not isolated abstractions, but are populated with contract-specific facts, entities, and relations that conform to the induced conceptual structure. In this way, the methodology separates conceptual stabilization from instance generation, en-

Figure 1: Semantic Onboarding methodology for constructing ontology-enriched formal knowledge models from contract artifacts.



abling more consistency across multiple onboarded contracts. The ABox generation phase is therefore central to transforming the formal schema into executable and contract-sensitive knowledge models.

A key distinguishing phase of the methodology is *ontology–ABox alignment*. At this point, the generated formal knowledge is aligned with one or more selected ontologies that provide domain-independent semantic categories and relations. In the context of contract management, this alignment is particularly important because it connects contract-specific formalizations to a broader ontological foundation, allowing the resulting models to be interpreted within a semantically principled framework. Ontology alignment is performed by mapping the extracted entities and relations to the corresponding concepts and predicates defined in the selected contract ontology, providing semantic grounding for the generated representations.

The methodology also includes a dedicated *ontology selection* activity. Since ontology alignment depends on the semantic scope of the contracts and the type of decision-support analysis being targeted, the choice of ontology is treated explicitly rather than implicitly. This phase allows the onboarding process to be guided by a selected ontological foundation appropriate to the domain, such as a contract-oriented ontology grounded in foundational categories. By making ontology selection explicit, the methodology acknowledges that semantic grounding is shaped not only by the source artifacts, but also by the conceptual commitments adopted during modeling.

Another essential element of the methodology is *human validation*, which interacts with multiple phases of the process. Human validation is not treated as a final correction step only, but as a transversal quality-control mechanism capable of intervening at critical points of model construction. Validation may occur over the initial formal models, the induced TBox, and the generated ABoxes, depending on the level of confidence, domain sensitivity, and modeling complexity involved. This human-in-the-loop component is especially relevant in contract domains, where subtle legal, operational, or institutional distinctions may have significant implications for the resulting knowledge structures. Consequently, human validation contributes to preserving semantic adequacy and modeling reliability throughout the onboarding process.

The final output of the methodology is a set of *ontology-enriched formal knowledge models*. These models constitute the result of the entire Semantic Onboarding process and represent the semantic substrate later used by the architecture presented in the previous section. Because they are formally structured, semantically aligned, and validated, these models provide a stronger grounding basis than raw textual fragments or isolated database records. They can therefore support semantically aware decision-support applications, including contract analytics, compliance analysis, risk assessment, and other reasoning-intensive tasks.

In general, the Semantic Onboarding methodology establishes a structured path from heterogeneous contractual artifacts to semantically grounded formal representations. Its main contribution lies in combining the flexibility of LLM-based formalization, the stability of deterministic schema construction, the semantic rigor of ontology alignment, and the reliability of human validation. This combination enables the construction of knowledge models that are not merely extracted from data, but semantically engineered for downstream intelligent use.

5 Architecture

Figure 2 presents the architecture of the proposed semantic-aware decision-support system for contract management. The architecture is organized around a central orchestration component, called *IntelliAgent*, which mediates the interaction between users, semantically grounded contract knowledge, and auxiliary

analytical capabilities. Unlike conventional LLM-based architectures that rely primarily on direct retrieval from document stores or structured databases, the proposed architecture is centered on a *Semantic Layer* that encapsulates formally constructed knowledge derived from contract artifacts. This design allows the system to support decision-making not only through data access, but through semantically grounded interpretation.

At the top of the architecture is the *User Query* component, which represents the entry point of the system. Users interact with the platform by posing natural-language questions related to contract analysis, managerial assessment, compliance verification, or broader decision-support needs. The connection between the user and IntelliAgent is bidirectional, reflecting an interactive process rather than a one-shot request-response pipeline. This bidirectionality is important because the system may refine interpretations, maintain contextual continuity, and adapt responses according to the analytical needs associated with the query.

The core orchestrator is the *IntelliAgent*. This component is responsible for interpreting the user request, managing the interaction with the semantic infrastructure, and coordinating auxiliary analytical services whenever additional processing is required. Architecturally, IntelliAgent does not operate as a direct retrieval agent over heterogeneous raw sources. Instead, it is positioned above the Semantic Layer and interacts with it as the primary knowledge substrate of the system. This design decision reflects the architectural shift proposed in this work: rather than grounding responses directly on retrieved data fragments, the system grounds them on a formal semantic representation previously constructed from contract artifacts.

The *Semantic Layer* is the central knowledge component of the architecture. It provides the semantically structured representation over which the system operates when analyzing user requests. From an architectural perspective, this layer acts as the bridge between raw contractual artifacts and intelligent analytical behavior. It encapsulates the formal concepts, relations, and interpretive structures necessary to support contract-oriented reasoning in a manner that is explicit, inspectable, and reusable across queries. Instead of exposing raw documents or isolated database tuples to the reasoning process, the Semantic Layer presents an organized knowledge substrate capable of supporting semantically coherent responses.

This layer is supported by four main internal components. The first is the *UFO Service Contract Ontology*, which provides the conceptual vocabulary used to organize contract knowledge. Architecturally, the ontology functions as the semantic backbone of the system, defining the categories, distinctions, and relationships through which contract elements can be represented in a formally meaningful way. Its role in the architecture is not merely terminological; it provides the structuring foundation that enables contract data to be interpreted according to a shared conceptual model.

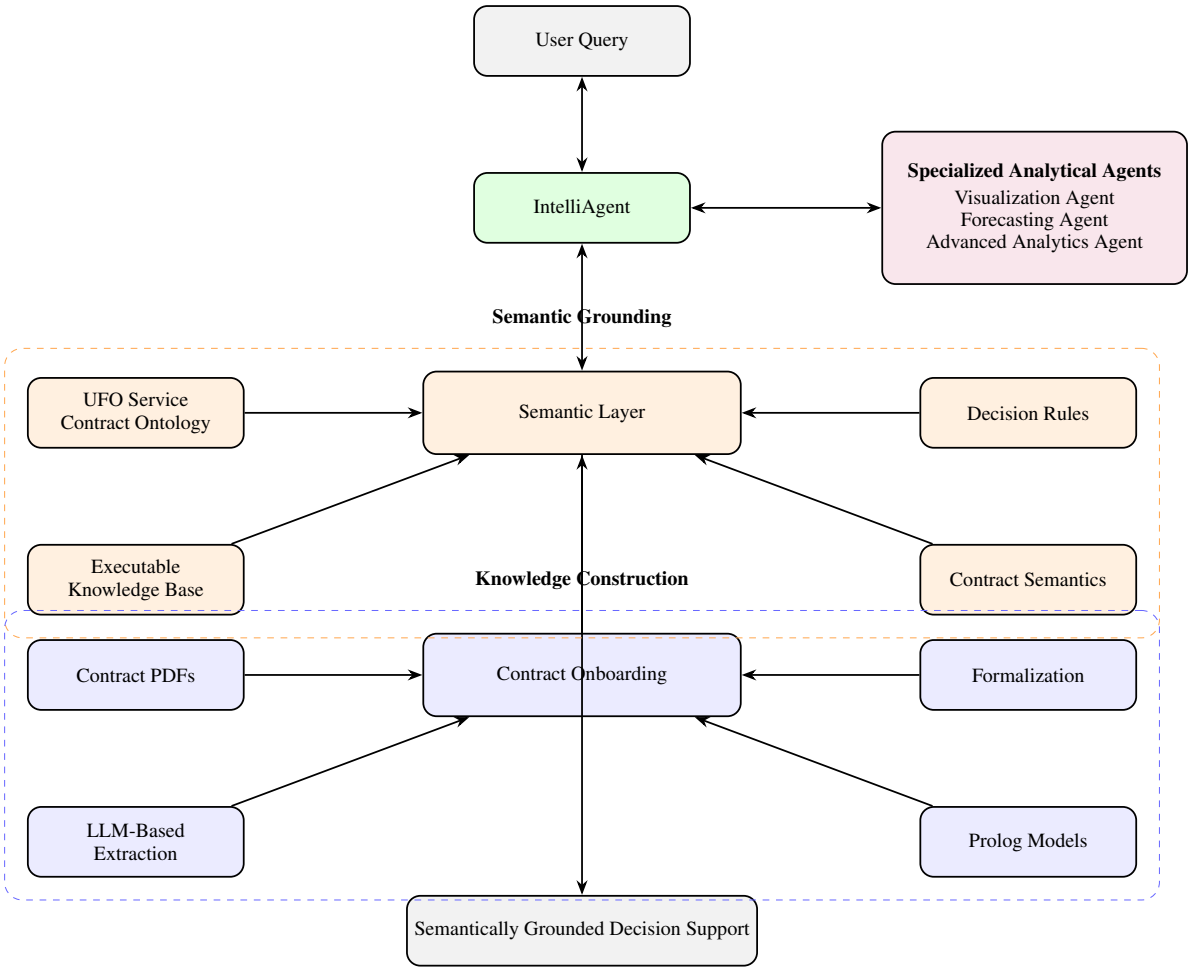
The second component is the *Executable Knowledge Base*. This component stores the formalized contract knowledge in a machine-interpretable and executable representation. Within the architecture, it serves as the operational repository of semantically grounded facts and structures. Because this knowledge base is executable, it can support computational access to contract semantics beyond simple lookup, enabling the system to rely on structured representations rather than only on textual retrieval.

The third component is *Decision Rules*. Architecturally, this component contains the explicit rules that support the transformation of semantically represented contract information into decision-relevant outcomes. These rules provide the system with a structured way to connect contract semantics to evaluative or analytical interpretations. Their inclusion in the architecture is important because it distinguishes the platform from systems that only retrieve or summarize information. Here, the architecture explicitly reserves a place for formal rule-based support to decision-oriented analysis.

The fourth component is *Contract Semantics*, which represents the semantically enriched interpretation of contract elements as a coherent whole. While the ontology provides the conceptual schema and the knowledge base stores executable representations, this component reflects the resulting semantic content that becomes available for use by the system. In architectural terms, it is the layer through which the meaning of contractual clauses, obligations, conditions, and other contractual structures becomes available to higher-level analytical processes.

Below the Semantic Layer lies the *Contract Onboarding* component, which is responsible for supplying the architecture with semantically processable contract knowledge. In the context of the architecture, this component should be understood as the entry point for incorporating contractual artifacts into the system. Its role is to prepare the informational basis from which the Semantic Layer is formed. Although the specific transformation process is methodological in nature and therefore discussed later, architecturally this component represents the subsystem dedicated to bringing external contract artifacts into the platform in a structured form.

Figure 2: Semantic-aware architecture for contract management decision support.



The Contract Onboarding component is itself connected to four supporting elements. The first is *Contract PDFs*, which represent the original documentary artifacts from which knowledge is derived. These are the primary source documents ingested by the system. The second is *LLM-Based Extraction*, which denotes the use of language-model capabilities to identify and extract relevant contractual content from those documents. At the architectural level, this component provides the mechanism through which unstructured textual artifacts become accessible to subsequent formal organization.

The third onboarding component is *Formalization*. This element represents the transition from extracted contractual content to formally structured representations. Its presence in the architecture emphasizes that the system does not stop at textual processing, but includes an explicit stage devoted to transforming extracted information into semantically organized knowledge artifacts. The fourth component is *Prolog Models*, which correspond to the formal representations generated from that process. Architecturally, these models constitute the concrete formal artifacts that feed the Semantic Layer and support executable access to contract knowledge.

At the right side of the architecture, Figure 2 also includes a set of *Specialized Analytical Agents*. These agents are not part of the main semantic grounding path, but they can be invoked by IntelliAgent whenever the user request requires additional analytical functionality. Their relation with IntelliAgent is bidirectional, indicating dynamic interaction and orchestration. Examples include a *Visualization Agent*, responsible for producing charts and graphical summaries; a *Forecasting Agent*, aimed at temporal or trend-based analyzes; and an *Advanced Analytics Agent*, which may support more specialized analytical operations. Their architectural role is complementary: they extend the system's analytical capabilities without replacing the Semantic Layer as the primary

grounding mechanism.

At the bottom of the architecture is the *Semantically Grounded Decision Support* component, which represents the final system output. This output is not conceived merely as a generated answer, but as a decision-support artifact grounded in the semantic structures maintained by the architecture. From an architectural standpoint, this component synthesizes the purpose of the system: to transform contract documents into a semantically structured knowledge substrate and to use that substrate as the basis for intelligent, context-aware, and decision-oriented support.

The architecture is implemented in Python for orchestration and data processing, uses Gemini 2.0 for knowledge extraction and formalization tasks, and employs SWI-Prolog as the executable reasoning engine underlying the Semantic Layer. Overall, its main distinguishing characteristic is that user interaction is mediated by a formal semantic layer built from contract artifacts, while still preserving extensibility through specialized analytical agents. This organization separates the concerns of knowledge construction, semantic representation, orchestration, and auxiliary analytics, thereby providing a modular and extensible foundation for contract-oriented intelligent systems.

6 Evaluation

The main advantage of the proposed approach becomes evident when evaluating semantically grounded queries that require interpreting contractual meaning rather than retrieving explicit textual fragments. Unlike the baseline system, which depends on contextual synthesis over retrieved text, the proposed architecture enables direct querying over formally structured and ontology-aligned knowledge.

To illustrate this capability, we consider the following question: *Which requests from the contracting party must be prioritized by the contractor, as established in the contract?*

Answering this question requires identifying the obligations of the contractor that correspond to enforceable expectations from the contracting party. In the proposed system, this mapping is explicitly represented through legal relations encoded in the ontology-aligned ABox.

From Contract Text to Semantic Representation The Semantic Onboarding process transforms contractual artifacts into structured and semantically grounded representations. Table 1 summarizes the main artifacts generated during this process, including schema definitions (TBox), instance-level representations (ABox), ontology-aligned models, and example queries and results.

Semantic Interpretation and Reasoning The table illustrates how contractual knowledge evolves from raw structured facts to semantically enriched representations. While the TBox defines the schema and the ABox captures contract-specific data, the ontology-aligned ABox introduces explicit semantic relations such as `duty_to_act` and `right_to_action`. These relations enable the system to represent contractual obligations and enforceable expectations in a structured and machine-interpretable form.

Unlike traditional approaches, where such relationships must be inferred from textual fragments, the proposed system encodes them explicitly. This enables direct querying of contractual semantics through logical predicates, allowing the system to aggregate obligations across clauses and identify enforceable actions in a systematic manner.

Answering Semantically Grounded Questions Using the representations shown in Table 1, the motivating question can be answered through logical queries that combine rights and duties. Specifically, actions that appear both as `right_to_action` for the contracting party and as `duty_to_act` for the contractor correspond to enforceable requests defined in the contract.

This approach contrasts with the baseline system, where answering such a question would require retrieving multiple textual fragments and relying on the LLM to infer relationships between them. In the absence of explicit semantic structure, this process is inherently approximate and may lead to incomplete or inconsistent results.

Comparison with the Baseline System In the baseline architecture, answering semantically complex questions requires retrieving multiple textual fragments and relying on the LLM to implicitly infer relationships between them. This process is inherently indirect, as the system operates over unstructured or weakly structured representations, without explicit modeling of contractual semantics such as obligations, rights, or enforcement relations. As a result, answers may vary depending on prompt design, retrieved context, and model interpretation, often leading to incomplete or non-systematic responses.

In contrast, the proposed system enables direct querying over formally structured and ontology-

Table 1: Formal artifacts and queries generated during Semantic Onboarding

Component	Prolog Representation

TBox	`class(contract).`
	`predicate_signature(contract,1).`
	`predicate_signature(contract_metadata,3).`
	`predicate_signature(contract_clause_fact,5).`

ABox	`contract(contract_002_2026).`
	`contract_metadata(contract_002_2026, contracting_party, a_bank).`
	`contract_metadata(contract_002_2026, contractor, d_tech).`
	`contract_clause_fact(contract_002_2026, clause_contract_guarantee,`
	`guarantee_amount, '133000.00', clause_text).`

Ontology-ABox	`plays_role(a_bank, service_customer_role, contract_002_2026).`
	`plays_role(d_tech, hired_service_provider_role, contract_002_2026).`
	
	`legal_relation_instance(clause_service_levels, duty_to_act,`
	`d_tech, execute_services_according_to_standards).`
	
	`legal_relation_instance(clause_service_levels, right_to_action,`
	`a_bank, require_service_execution_according_to_standards).`

Query (Duties)	`?- legal_relation_instance(_, duty_to_act, d_tech, Action).`

Result (Duties)	`Action = execute_services_according_to_standards.`
	`Action = provide_contractual_guarantee.`
	`Action = maintain_confidentiality_of_information.`
	`...`

Query (Rights + Duties)	`?- legal_relation_instance(_, right_to_action, a_bank, Action),` `legal_relation_instance(_, duty_to_act, d_tech, Action).`

Result (Enforceable Actions)	`Action = execute_services_according_to_standards.`
	`Action = maintain_confidentiality_of_information.`
	`...`

aligned knowledge. By representing contractual semantics explicitly, the system can retrieve obligations, rights, and related actions in a consistent and traceable manner. This difference becomes particularly evident when analyzing semantically grounded questions, as illustrated in Table 2.

As shown in Table 2, the baseline system performs adequately for factual and descriptive queries, where answers can be directly retrieved from documents or structured databases. However, its limitations become evident for questions that require semantic interpretation, aggregation across clauses, or identification of implicit relationships. In these cases, the absence of explicit semantic structure forces the LLM to approximate reasoning through textual synthesis.

To complement the illustrative example, both approaches were also applied to ten real-world IT service contracts using the competency question *What is the cybersecurity criticality level of the contract?*. The expected classification for each contract was manually established according to the organizational criticality criteria. The baseline approach produced classifications consistent with the expert assessment in six contracts, whereas the proposed semantic-aware approach was consistent with the expert assessment in all ten evaluated contracts.

Semantic Completeness Metric To further characterize the differences between the two approaches, we introduce the notion of *semantic completeness*, defined as the extent to which a system is able to identify and return all relevant elements required to answer a semantically grounded question. Let Q be a semantic query (e.g., obligations of a contractor), and let A_{true} be the set of all relevant actions present in the contract. Let A_{system} be the set of actions returned by the system. This notion captures not only retrieval coverage but also the system's ability to correctly interpret and aggregate semantically related elements across distributed contract clauses. We define semantic completeness as:

$$\text{SemanticCompleteness}(Q) = \frac{|A_{system} \cap A_{true}|}{|A_{true}|} \quad (1)$$

Table 2: Comparison of question-answering capabilities between the baseline system and the proposed semantic-aware approach.

Question Type	Baseline System (RAG + SQL)	Proposed Semantic System
What is the total value of the contract?	Directly retrieved from structured data or documents.	Directly retrieved from formal representation.
What is the deadline for contract extension?	Retrieved from clause fragments via RAG.	Retrieved from structured facts in the ABox.
Which obligations does the contractor have?	Requires aggregation of multiple text fragments; implicit inference by LLM.	Directly retrieved via `duty_to_act` predicates.
Which actions can the contracting party enforce?	Not explicitly represented; inferred heuristically.	Directly retrieved via `right_to_action` predicates.
Which requests from the contracting party must be prioritized by the contractor?	Indirect and potentially incomplete; depends on prompt and retrieval quality.	Directly computed by combining `right_to_action` and `duty_to_act`.
What penalties can be applied in case of non-compliance?	Retrieved as isolated clauses; relationships must be inferred.	Explicitly represented via `power` and `subjection` relations.

This metric captures how completely a system is able to retrieve semantically relevant information. Table 3 provides a qualitative comparison of the expected behavior of both systems under this metric. The baseline system does not guarantee semantic completeness, as it relies on the retrieval of relevant fragments and subsequent interpretation by the LLM. Missing or weakly ranked fragments may lead to incomplete answers. In contrast, the proposed system operates over a structured knowledge base, allowing it to systematically enumerate all relevant elements associated with a given semantic relation.

In summary, the comparison demonstrates that the introduction of a semantic layer fundamentally changes the nature of query answering. While the baseline approach is effective for data retrieval, the proposed system enables semantically grounded reasoning, improving completeness, consistency, and explainability in contract-oriented decision support. All experimental artifacts used in this study, including source code, datasets, ontology models, Prolog knowledge bases, generated outputs, prompts, and evaluation logs, are publicly available in our Github repository (Seabra, 2026).

7 Conclusions and Future Works

This paper introduced a semantic-aware architecture for Business Intelligence in contract management, grounded on ontology-enriched formal knowledge models constructed through a Semantic Onboarding methodology. By introducing a Semantic Layer built upon formal representations aligned with a domain ontology, the proposed approach shifts LLM-based BI from retrieval-oriented data integration toward semantically grounded reasoning. The evaluation demonstrated that ontology-aligned formal representations enable direct querying of contractual obligations, rights, and enforcement relations, providing more consistent, complete, and explainable answers to semantically grounded questions. These results suggest that semantic grounding can serve as an effective foundation for contract-oriented decision-support systems.

Beyond the contract management domain, the proposed approach illustrates how semantic engineering can complement LLM-based analytics in scenarios where decision-relevant knowledge is distributed across heterogeneous artifacts and cannot be reliably recovered through retrieval alone. By explicitly representing domain semantics and decision criteria, the architecture provides a foundation for more transparent and auditable decision-support processes.

Despite these contributions, the approach remains subject to limitations associated with ambiguous contractual language, ontology coverage gaps, and potential inaccuracies introduced during LLM-assisted extraction and formalization, making human validation an important quality-control mechanism. Future work will investigate schema evolution mechanisms for incrementally adapting the TBox as new contracts are incorporated, support for open-world assumptions in situations involving incomplete information, and the integration of additional ontological dimensions such as risk, compliance, and value. We also plan to explore tighter neuro-symbolic integration strategies and to evaluate the proposed approach in other domains characterized by heterogeneous information sources and implicit semantics.

Table 3: Qualitative comparison of semantic completeness.

Approach	Expected Semantic Completeness
Baseline (RAG + SQL)	Partial. Depends on retrieval coverage and prompt interpretation. May omit obligations distributed across clauses or expressed in different forms.
Proposed Semantic System	High. Retrieves all relevant actions through formal predicates (e.g., `duty_to_act`, `right_to_action`), ensuring systematic coverage of the contract.

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